

High Efficiency Breather Filters

Application Maintenance Site Management Component Rebuild Safety MARC Management

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1.0 Introduction

A compartment breather filter plays a critical role in extending component life. The breather filter must allow air to easily pass through it while preventing microscopic particulate particles from entering the compartment and contaminating the fluid with abrasive debris.

Fuel, hydraulic and powertrain systems use a wide variety of breather filters, which means a mine must stock multiple filters to accommodate all machines in its fleet. There are no specific design guidelines for the size, type, or efficiency of breather filters.

2.0 Best Practice Description

On newer machines, increasingly sophisticated, high-performance systems and components require very clean fluids. Most standard breather filters are not efficient enough to provide maximum protection to new systems. In addition, a mine must stock multiple part numbers of standard breathers to service a fleet.

An alternative is to standardize on the use of spin-on High Efficiency 4-micron fuel filters as breathers for all compartments. This includes fuel, hydraulic, transmission, etc. These filters are very high performance and reduce the number of breather part numbers that have to be stocked. A common practice is to use the same HE fuel filter as is used on one or more of the machines in the fleet.

3.0 Implementation Steps

When used as breather filters, HE fuel filters are mounted upside down so that falling dirt and debris are deflected from the top of the filter. The filter is mounted on a combination mounting stud and mounting bracket that is fabricated by the dealer. A hose is then installed between the mounting bracket and compartment. Care must be taken to insure that only non-vented compartment caps are used so that all air entering and leaving the compartment must pass through the breather filter.

4.0 Benefits

Dust entering fluid compartments through inefficient breather filters is often a source of fluid contamination. This can be easily prevented with the use of HE fuel filters as breathers. When used as fuel tank breathers, HE filters have reduced or eliminated premature fuel filter plugging in extremely dusty applications. HE fuel filters are also larger than standard breathers and have much greater capacity. There have been no documented cases of these filters plugging. They are typically replaced only once or twice a year.

5.0 Resources Required

Mounting brackets and hoses must be fabricated and installed by the dealer. This requires very simple machining, welding and hose-making capabilities. See diagram contained [Component Life Management Master Document](#) PDF (click on attachments in this file).

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6.0 Supporting Attachments

[Component Life Management Master Document](#) PDF (click on attachments in this file)

7.0 Related Best Practices

0806-2.10-1000 -Managing Fluid Cleanliness

8.0 Acknowledgements

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